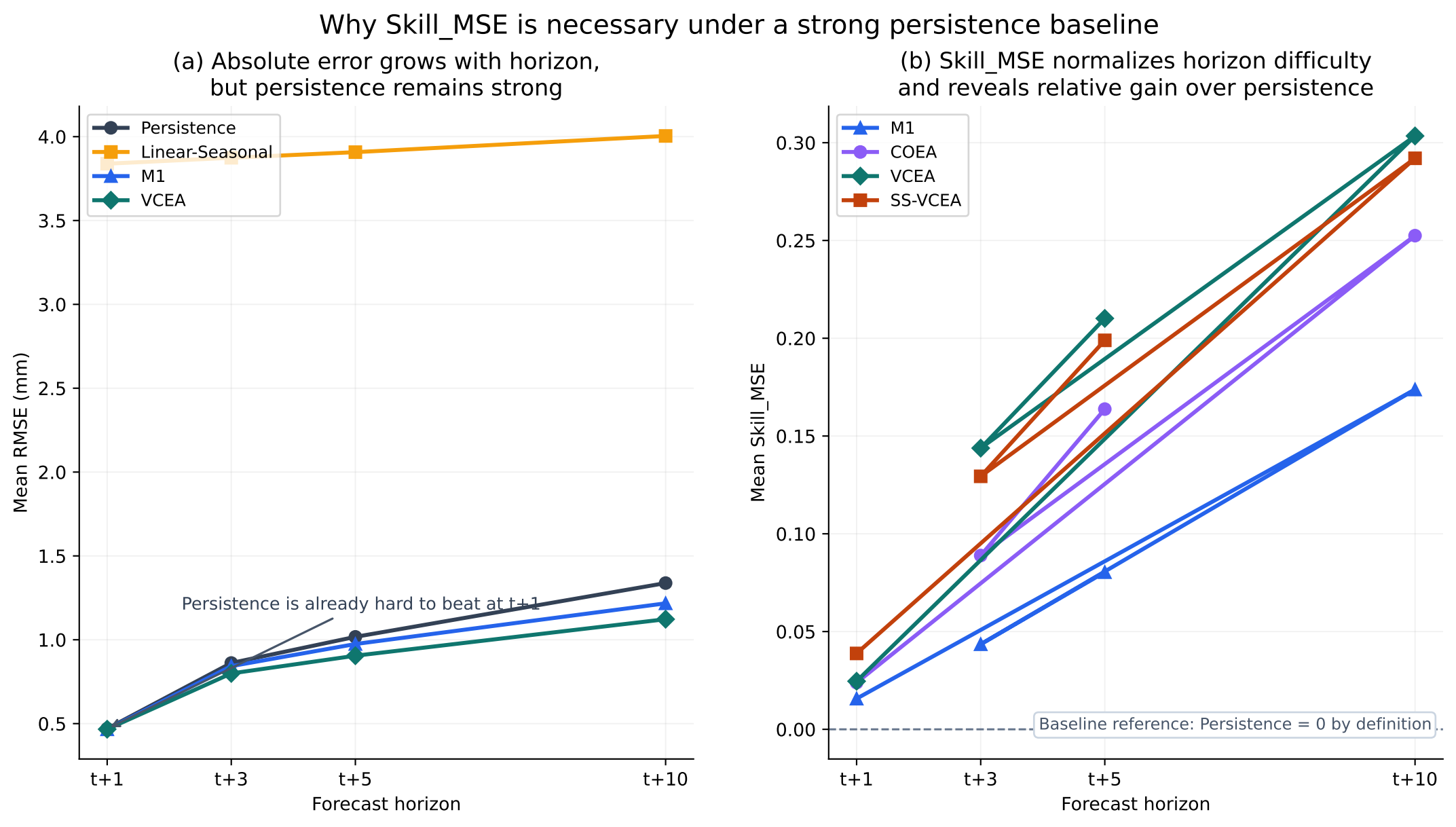
Why Skill_MSE is necessary under a strong persistence baseline
(a) Absolute error grows with horizon,
but persistence remains strong
Persistence
Linear-Seasonal
M1
VCEA
Mean RMSE (mm)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
Persistence is already hard to beat at t+1
t+1
t+3
t+5
t+10
Forecast horizon
(b) Skill_MSE normalizes horizon difficulty
and reveals relative gain over persistence
M1
COEA
VCEA
SS-VCEA
Mean Skill_MSE
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Baseline reference: Persistence = 0 by definition
t+1
t+3
t+5
t+10
Forecast horizon